

■ What Is a Ratio?

Comparing Things Around Us!

Subject:	Mathematics — Numbers & Operations
Duration:	45 Minutes
Topic:	Introduction to Ratios
Context:	Culturally responsive — Indian classroom setting

■ Learning Objectives

By the end of this lesson, students will be able to:

- ✓ Define ratio and explain what it means to compare two quantities.
- ✓ Write a ratio in the correct **a : b** notation.
- ✓ Recognise that order matters in a ratio ($3:1 \neq 1:3$).
- ✓ Identify real-life situations where ratios are used.
- ✓ Solve simple ratio problems set in familiar, everyday contexts.

■ Materials Required

- Whiteboard and markers
- Coloured chalk or counters (red and yellow)
- Small bowls with objects (e.g. gulab jamuns and barfi cutouts)
- Worksheet printouts (Exit Ticket)
- Optional: projected slides from the source presentation

Lesson Flow at a Glance

Phase	Activity	Time
1. Hook	The Snack Box Puzzle — spark curiosity	5 min
2. Introduce	Priya's Rangoli Problem — real-life context	5 min
3. Teach	What is a Ratio? — definition + notation	8 min
4. Examples	Chai Recipe, Classroom, Sweet Box Ratios	7 min
5. Key Rule	Order Matters! ($3:1 \neq 1:3$)	5 min
6. Practice	Guided Practice 1 & 2 — students try together	8 min
7. Connect	Where Is Ratio Hiding? — real world links	4 min
8. Review	Vocabulary Wall + 3-Point Recap	3 min
9. Exit Ticket	3 quick questions — individual check	5 min

Phase 1 — Hook: The Snack Box Puzzle (5 min)

Teacher action: Hold up (or draw) a box with **4 samosas** and **2 laddoos**.

Ask the class:

"Which snack is more? By how much?"

"Can we write this as a comparison?"

Expected student responses: Samosas are more. There are 2 extra samosas. We have twice as many.

Bridge: 'Today we will learn a neat mathematical way to write exactly this kind of comparison — it is called a **RATIO**.'

Phase 2 — Real-Life Context: Priya's Rangoli (5 min)

Tell the story:

Priya is making a beautiful rangoli for Diwali. She has **3 pots of red powder** and **1 pot of yellow powder**. How can she describe what she has?

Elicit ideas from students, then guide them:

→ *She could say: 'I have 3 reds and 1 yellow.'*

→ *In maths we write this as: **3 : 1** (read as 'three to one').*

Phase 3 — What Is a Ratio? (8 min)

Write the definition on the board:

A **ratio** is a way of **COMPARING** two quantities.
We write it as **a : b** (read as 'a to b').

Phase 4 — Worked Examples (7 min)

Example 1 — Chai Recipe:

2 cups milk : 1 cup water $\rightarrow$ ratio = **2 : 1**

Example 2 — Classroom (dal : rice recipe):

2 cups rice : 1 cup dal $\rightarrow$ ratio = **2 : 1**

Example 3 — Sweet Box:

3 gulab jamuns : 2 barfis $\rightarrow$ ratio = **3 : 2**

Example 4 — Classroom count:

4 girls : 2 boys $\rightarrow$ ratio = **4 : 2**

Phase 5 — Key Rule: Order Matters! (5 min)

Use Priya's rangoli colours to demonstrate:

Red to Yellow	Yellow to Red
3 : 1	1 : 3
3 times as much red	3 times as much yellow

■ Always write the ratio in the same order as the question asks. If asked 'red to yellow', write red first!

Phase 6 — Guided Practice (8 min)

Work through each question as a class. Call on students to answer.

Practice Set A — Write the ratio:

1. 5 girls and 3 boys → girls to boys

Answer: 5 : 3

2. 4 mangoes and 2 bananas → mangoes to bananas

Answer: 4 : 2

3. India scored 5 wickets, Pakistan scored 3 → India to Pakistan

Answer: 5 : 3

Practice Set B — What does the ratio mean?

4. A bag has 6 red marbles and 4 blue marbles. Write the ratio of blue to red.

Answer: 4 : 6 (blue first because the question asked blue to red)

5. True or False: 6 : 4 is the same as 4 : 6?

Answer: FALSE — order changes the meaning!

Phase 7 — Where Is Ratio Hiding? (4 min)

Invite students to think of their own examples. Offer prompts:

Location	Example Ratio
Kitchen ■	Dal to rice (1 : 2)
Road traffic ■	Cars : bikes : autos (multi-term ratio)
Vegetable market ■	Tomatoes : potatoes (3 : 5)
Cricket match ■	Runs by team A : runs by team B
Classroom ■	Boys : girls in your class
Flower pot ■	Red flowers : yellow flowers

Phase 8 — Vocabulary Wall & 3-Point Recap (3 min)

Term	Definition
RATIO	A way of comparing two quantities using the notation a : b

COMPARE	Looking at the difference or relationship between two things
QUANTITY	How many or how much of something there is
ORDER	The sequence in which the two quantities appear in the ratio

3-Point Recap | 1. A ratio COMPARES two things. 2. We write it as a : b. 3. ORDER matters — 3:1 $\neq$ 1:3.

⇒ ■ Exit Ticket — Individual Assessment (5 min)

Students answer independently on the slip below (or in their notebook).

Q	Question	Answer
1	Write the ratio of 4 red marbles to 2 blue marbles.	4 : 2
2	True or False — 2:3 is the same as 3:2?	FALSE
3	5 apples and 3 oranges — write the ratio of apples to oranges.	5 : 3

■ Differentiation & Extension

Support (students who need more help):

- Use physical objects (coins, counters, coloured blocks) to count before writing.
- Provide a ratio sentence frame: 'There are ____ [A] for every ____ [B], so the ratio is ____:____'

Extension (students who finish early):

- What if the ratio is 4:2? Can you simplify it? (Answer: 2:1)
- Make up your own story problem that uses a ratio and challenge a partner to solve it.
- Look around the classroom — write 5 ratios you can spot right now.

■ Assessment & Observation Notes

Formative:

- Observe responses during guided practice (whiteboards or verbal).
- Check exit ticket — students scoring 2/3 or below need follow-up.

Common misconceptions to watch for:

- Writing ratio as addition (e.g. '3 + 1') instead of 3:1.
- Reversing the order (writing 1:3 when asked for red to yellow).
- Confusing ratio with fraction — clarify that 3:1 is not the same as $\frac{3}{1}$.

■ Homework / Follow-Up

Ask students to bring one **real-life ratio from home** to share the next day.

Examples they can look for:

- Ingredients in a recipe their family uses
- Number of windows to doors in their house
- Number of spoons to forks in the cutlery drawer
- Pages read vs pages remaining in a book

Bonus: Write the ratio in **a : b** form and write a sentence describing it.